# Appendix A - Survey

Q1. Are you currently working as a software practitioner?

- Yes (continue)
- No (terminate)

Q2. Are you currently using AI for your work as a software practitioner?

- Yes (continue)
- No (terminate)

Q3. How do you identify yourself?

- ☐ Man
- ☐ Woman
- ☐ Non-binary / gender diverse
- ☐ Prefer not to answer
- ☐ Prefer to self-describe as: _____

Q4. What is your age group?

- 18–24
- 25–34
- 35–44
- 45–54
- 55–64
- 65+
- Prefer not to say

Q5. In which country do you work? (primary physical location)

Q6. Years of experience

- 0–2 years
- 2–5 years
- 5-10 years
- 10-15 years
- 15-20 years
- 20-30 years
- 30-40 years
- 40-50 years
- 50+ years

Q7. Primary role

- Backend developer
- Frontend developer
- Full stack developer
- ML/AI developer
- Tester
- DevOps
- Project/Product/Program Manager
- Business Analyst
- Software Architect
- Product Owner
- Others (please specify): _____

Q8. What are the main job responsibilities of your primary role?

Q9. What is the primary industry or domain of the company you work for (e.g., finance, automotive, healthcare, education, retail, telecommunications, government, entertainment, manufacturing, technology/software)?

Q10. What is the nature of the organisation you are affiliated with in your primary role?

- I am Self-Employed
- Extra Small (2-10 employees)
- Small (10-100 employees)
- Medium (101 - 500 employees)
- Large (501 - 10,000 employees)
- Extra Large (more than 10,000 employees)

Q11. How many team members are part of your primary team or the team you are most contributing to?

- No other members
- 2 - 5
- 6 - 10
- 11 - 20
- More than 20

Q12**.** How much do you agree or disagree with the following statements? (5-point Likert: Strongly disagree → Strongly agree)

1. Using AI tools improves my effectiveness as a software practitioner.
2. I feel comfortable incorporating AI-generated outputs in my software work.
3. I prefer to use AI assistance even when I could complete a task myself.
4. I am interested in experimenting with new AI tools in my workflow.
5. I am concerned about the long-term impact of AI on my role.

6. My use of AI tools is primarily driven by curiosity and personal interest.
7. I use AI tools because I am expected to by my employer or team.

**CURRENT PRACTICE (AI as X):**

The following questions are about your current experience of working with AI in software engineering tasks.
People may work with AI in different ways depending on the task, team, or tool being used. The scenarios below are not predictions or ideal states, they are examples of possible ways AI may be involved in your work today.

**Q13.** Which of the following **best** describes how you currently work with AI **most of the time**?

- **Working Mode A: AI as a Support Tool**
  You carry out most of your own work by yourself. AI assists with small suggestions, documentation, quick fixes in routine SE tasks when needed.
- **Working Mode B: AI as an Assistant**
  You describe what you want in the SE task you are working on, and AI generates drafts, options, or explanations. You refine and adapt its output to fit your needs.
- **Working Mode C: AI as a Co-worker**
  You work in a continuous back-and-forth with AI on a given SE task, jointly shaping work outputs and decisions.
- **Working Mode D: AI as an Agent**
  You assign an SE task to an AI agent, and it works through the necessary steps to complete it and returns results, for your reference, review, or approval.
- **Working Mode E: AI as an Agentic System**
  You work with a system of AI agents that collectively carry out, coordinate among themselves, and maintain SE tasks.

**Q14.** How frequently do you use AI tools in your work?

- Multiple times per hour
- Multiple times per day
- Once a day
- A few times per week
- Rarely
- Never

**Q15.** In which **SE activities** do you currently **use AI most often at work**? (Select all that apply)

- ☐ Eliciting requirements
- ☐ Writing specifications
- ☐ Writing code
- ☐ Debugging

- ☐ Creating testcases
- ☐ Running testcases
- ☐ Technical documentation
- ☐ System design/architecture
- ☐ Code review
- ☐ DevOps/infrastructure
- ☐ Project management
- ☐ Team communication
- ☐ Other (please specify): _____

**Q16.** Select all the **AI tools you frequently use at work**? (Select all that apply)

- ☐ ChatGPT
- ☐ Claude
- ☐ Gemini
- ☐ GitHub Copilot
- ☐ Perplexity
- ☐ Cursor
- ☐ Windsurf
- ☐ Atlassian Rovo
- ☐ JetBrains AI Assistant
- ☐ Cline
- ☐ Lovable
- ☐ Amazon Q Developer
- ☐ Continue
- ☐ Aider
- ☐ Bolt.new
- ☐ Replit AI
- ☐ Other (please specify): _____
- ☐ I do not regularly use AI tools for work

**Q17.** Thinking about the **AI tool you use most frequently for work**, which interaction modes do you regularly use? (Select all that apply)

- ☐ Chat/conversation mode
- ☐ Inline code completion or suggestions
- ☐ Agent/autonomous task execution mode
- ☐ Command-line or terminal-based interaction
- ☐ Voice interaction
- ☐ Workflow automation/integration mode
- ☐ API-based usage
- ☐ Other (please specify): _____

**Q18.** How is **access to the AI tool you use most frequently** currently supported?

- Paid for by my employer
- I pay for access
- Free version only
- Other (please specify): _____

**Q19**. How does your **current experience of working with AI feel to you? (select all that apply)**

- ☐ Exciting
- ☐ Empowering
- ☐ Productive
- ☐ Uncertain
- ☐ Stressful
- ☐ Limiting
- ☐ Overwhelming
- ☐ Unrealistic
- ☐ Neutral
- ☐ Other (Please specify): _____

**Q20.** What level of **oversight** do you currently have when working with AI systems?

- **Oversight Level 1: Full Oversight**
  You review and approve all AI actions pertaining to your role before they take effect. You define the goals, set the boundaries, and make the final call on all decisions.
- **Oversight Level 2: Shared Oversight**
  You and the AI share responsibility over the workflow pertaining to your role. You handle some tasks and decisions independently while the AI does the same on others. You stay informed of what the AI is doing but do not need to approve every action. Control shifts between you and the AI depending on the task.
- **Oversight Level 3: Minimal Oversight**
  You set high-level goals and constraints over the workflow pertaining to your role. AI handles most of the decisions pertaining to your role, determines how tasks are approached, and drives the workflow forward. You step in only when something requires human judgment or approval or if something goes wrong.
- **Not applicable / I do not use AI systems with autonomous decision-making**

**FUTURES (AI as X)**

Below are several possible futures of working with AI in the software industry, which may already be in action for you currently.
These are not predictions. We are interested in knowing how you imagine and feel about them.

Q21. Which of the following futures best describes how you are most likely to work with AI in 2-3 years?

- **Future A: AI as a Support Tool**
  You carry out most of your own work by yourself. AI assists with small suggestions, documentation, quick fixes in routine SE tasks when needed.
- **Future B: AI as an Assistant**
  You describe what you want in the SE task you are working on, and AI generates drafts, options, or explanations. You refine and adapt its output to fit your needs.
- **Future C: AI as a Co-worker**
  You work in a continuous back-and-forth with AI on a given SE task, jointly shaping work outputs and decisions.
- **Future D: AI as an Agent**
  You assign an SE task to an AI agent, and it works through the necessary steps to complete it and returns results, for your reference, review, or approval.
- **Future E: AI as an Agentic System**
  You work with a system of AI agents that collectively carry out, coordinate among themselves, and maintain SE tasks.
- **Future F: AI as a Supervisor**
  AI assigns, prioritises, and evaluates how you carry out your software work.

(only for the selected future)
**BRANCHING QUESTIONS [Q12-22]**

**Q22.** How does this future **feel** to you? **(select all that apply)**

- ☐ Exciting
- ☐ Empowering
- ☐ Productive
- ☐ Uncertain
- ☐ Stressful
- ☐ Limiting
- ☐ Overwhelming
- ☐ Unrealistic
- ☐ Neutral
- ☐ Other (Please specify): _____

**Q23.** Thinking about this future, how do you expect your current role to change?

- It will no longer exist
- It will change significantly
- It will change moderately
- It will change slightly
- It will remain the same

[Conditional] > display only if any of the answer options 1-4 is selected

**Q24.** Please expand on why and how you expect your current role to change.

**Q25.** Thinking about this future, what would your role primarily involve? (select all that apply)

- ☐ Writing code
- ☐ Designing systems
- ☐ Reviewing AI output
- ☐ Coordinating AI systems
- ☐ Validating and ensuring quality
- ☐ Communicating with stakeholders
- ☐ Prompting/specification
- ☐ AI governance
- ☐ AI monitoring
- ☐ Verification/auditing
- ☐ Problem framing
- ☐ Strategic planning
- ☐ Other (please specify): _____

**Q26.** What is your desired level of **oversight of AI** in this future?

- **Oversight Level 1: Full Oversight**
  You review and approve all AI actions pertaining to your role before they take effect. You define the goals, set the boundaries, and make the final call on all decisions.
- **Oversight Level 2: Shared Oversight**
  You and the AI share responsibility over the workflow pertaining to your role. You handle some tasks and decisions independently while the AI does the same on others. You stay informed of what the AI is doing but do not need to approve every action. Control shifts between you and the AI depending on the task.
- **Oversight Level 3: Minimal Oversight**
  You set high-level goals and constraints over the workflow pertaining to your role. AI handles most of the decisions pertaining to your role, determines how tasks are approached, and drives the workflow forward. You step in only when something requires human judgment or approval or if something goes wrong.
- **Oversight Level 4: No Human Oversight**
  You define little to no direction over the workflow pertaining to your role. AI independently determines goals, makes decisions, carries out tasks, and manages the workflow without requiring your input, review, or approval. You are minimally involved or not involved at all in day-to-day decision-making and task execution.

**Q29.** Is there anything else you would like to share with us?